

IMU-Based Teeth-Click Gesture Feasibility Study Using PhyPhox Application

*From sensor data acquisition to Python based preprocessing,
visualization and analysis*

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Abstract

This document contains an end-to-end workflow followed for acquiring and analyzing inertial sensor data to explore teeth-click micro-gestures as a potential hands-free control input for hearables/headphones. Using the phyphox smartphone application, accelerometer and gyroscope signals were recorded during multiple trials of single, double, and triple teeth clicks under different motion and noise conditions (e.g., coughing and chewing). For prototyping, the smartphone was worn near the ear/temple region, i.e., in the vicinity of the temporomandibular joint (TMJ), to improve coupling between jaw activity and the IMU. The exported datasets were trimmed to remove noise from start/stop artifacts, consolidated into a unified IMU table, and analyzed in Python through required time-domain plots and basic statistics. Initial efforts were also made to differentiate the actual click from the noise generated during coughing and chewing. Future work is projected to differentiate and validate the single, double, and triple clicks by analyzing the actual time difference between the impulses generated.

Keywords: IMU, accelerometer, gyroscope, Phyphox, TMJ, teeth-click, signal processing, feature engineering

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1. Introduction

Have you ever been stopped by a stranger or a ticket inspector while both of your hands are busy with luggage and your ears are occupied with loud music? In that situation, interacting with a phone or earbud controls can be inconvenient or impossible. Modern hearables address hands-free control with head-motion gestures, but exaggerated nods or sudden head turns can feel socially awkward—especially in public, during a short interaction.

This motivates exploring subtle, discreet gestures that can be performed without freeing the hands or making noticeable movements. One candidate is soft teeth clicks / jaw micro-gestures, which are quick, repeatable, and can be performed with minimal visible motion. If such patterns can be reliably detected using inertial sensors (accelerometer/gyroscope) near the ear or temple, they could enable simple controls like pause/play, volume down, or acknowledge/answer- improving usability in real-world “hands and ears occupied” scenarios.

Therefore, this experiment evaluates whether a phone IMU can capture distinguishable signatures for single/double/triple teeth-click patterns under different head poses and motions.

2. Inertial Measurement Unit Background

An Inertial Measurement Unit (IMU) is a sensor module used to measure motion. In most consumer devices (phones, wearables, hearables), the IMU is built using MEMS technology (Micro-Electro-Mechanical Systems), where microscopic mechanical structures are fabricated on silicon and integrated with electronics.

A typical IMU contains at least a 3-axis accelerometer and a 3-axis gyroscope.

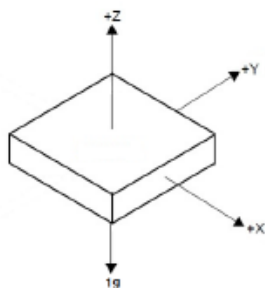


Figure 1 : 3-axis accelerometer

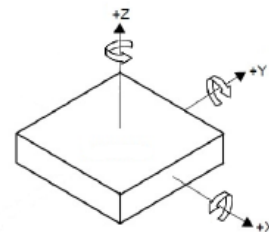


Figure 2 : 3-axis gyroscope

2.1 MEMS accelerometer

The accelerometer measures linear acceleration along x, y, and z (including the effect of gravity), and thus, the z-axis signal of the MEMS accelerometer is 1g (where $g = 9.81\text{m/s}^2$); while the x- and y-axis signals are zero in the ideal case without movement.

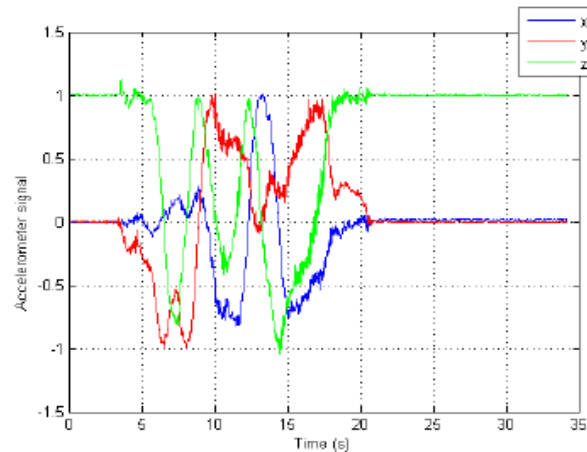


Figure 3 : Random accelerator signal

2.2 MEMS gyroscope

The MEMS gyroscopes is based on the Coriolis effect which measures angular velocity (rotation rate) around x, y, and z. All the three-axis signals are zero in the ideal case when it is stationary. When it is moved, the MEMS gyroscope detects the angular rate and the corresponding signal is in the unit of radians per second [rad/s].

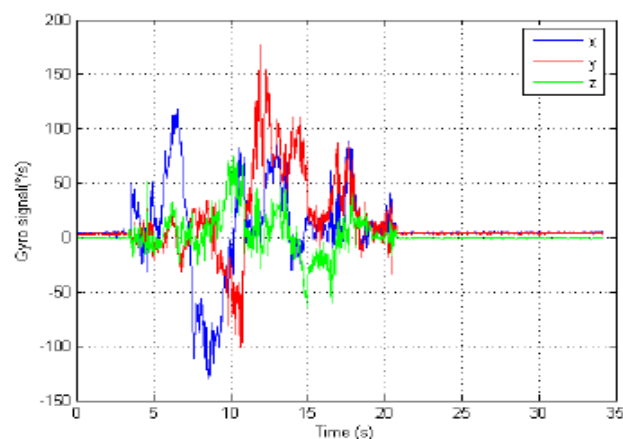


Figure 4 : Random gyroscope signal

3. Data Acquisition

To detect teeth-click patterns using inertial sensors, the first question is where the sensor should be placed so the motion caused by a click is observable. A teeth click is a short mechanical event that creates a small vibration and micro-movement around the jaw area, especially near the temporomandibular joint (TMJ). In real-world hearables/headphones, the device sits around the ear and presses against nearby skin and bone, which is physically close to the same region where these vibrations propagate. Based on this idea, the experiment places a smartphone (with built-in accelerometer and gyroscope) close to the TMJ area, aiming to capture click signatures as short, repeatable impulses in acceleration and angular-rate signals.

The goal of this stage is not perfect medical measurement of TMJ acoustics, but a practical confirmation that IMU signals contain patterns that can later be converted into a simple event-detection logic for single, double, and triple teeth clicks.

3.1 Physical setup for experiment

- The phone was positioned close to the TMJ region to capture micro-vibrations / small head-skin motions generated by teeth-click impulses. This placement is also conceptually relevant to hearables/headphones, because they physically contact nearby regions around the ear/temple.

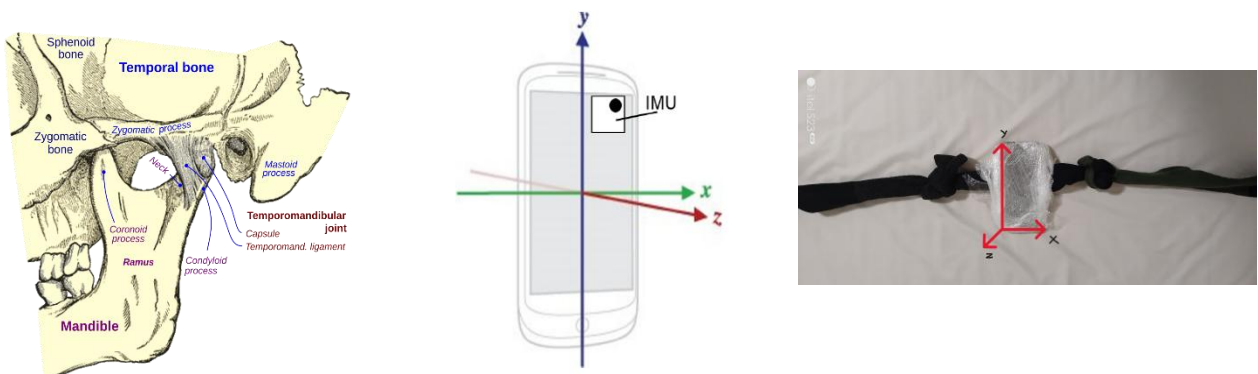


Figure 5 a. Temporomandibular Joint, b. IMU sensor in mobile, c. Physical setup for experiment

- The device was kept in portrait orientation, and the same side/position was maintained across trials for repeatability
- The phone is wrapped on the piece of fabric with a plastic roll and the fabric was used to hold the phone firmly against the TMJ area to ensure stable coupling and reduce slippage.

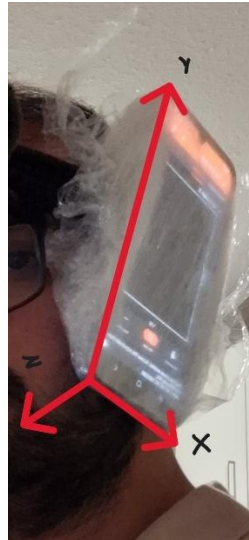


Figure 6 : Mounting IMU near test region

- Tightness was adjusted so the phone was secure with minimal wobble, but still comfortable.
- Trials were repeated with the same placement side and same orientation to keep signals comparable.
- *Any trial with visible repositioning or slipping of the phone was considered noisy, since mounting changes affect the measured IMU response.

3.2 Phyphox application configuration

- In phyphox app an experiment was configured to record both Accelerometer (X/Y/Z) and Gyroscope (X/Y/Z) simultaneously (raw sensor streams)

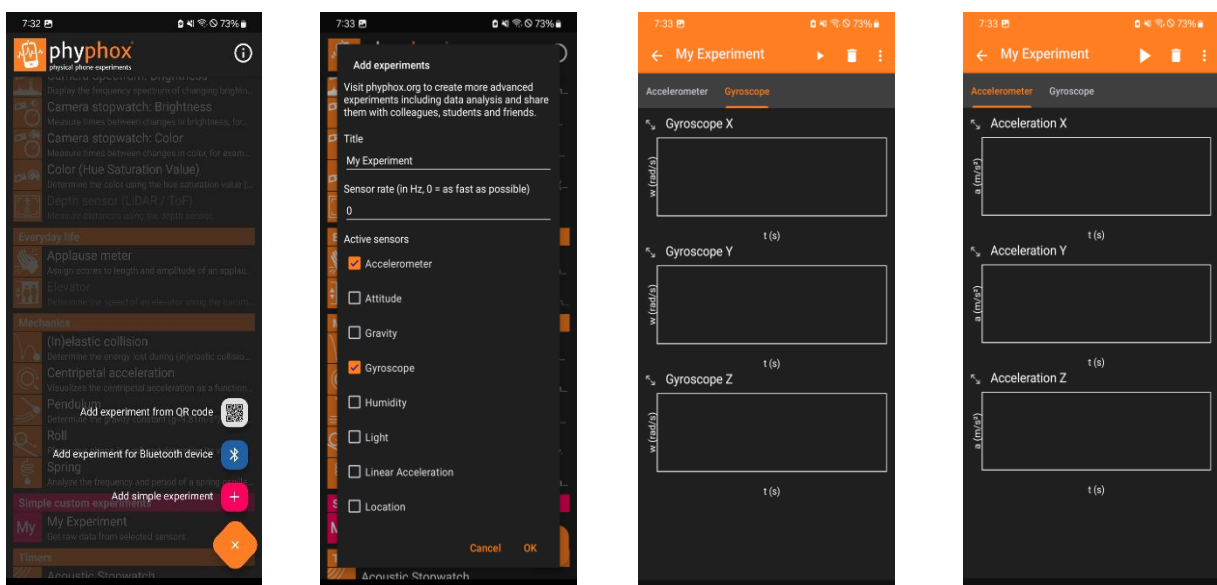


Figure 7 : Creating new experiment with required sensors

- The sampling rate was set to the maximum available (“fastest / max frequency”) to better capture short teeth-click impulses (limited by phone hardware; typically, a few hundred Hz). Set as 0 and select the required sensors as shown above.

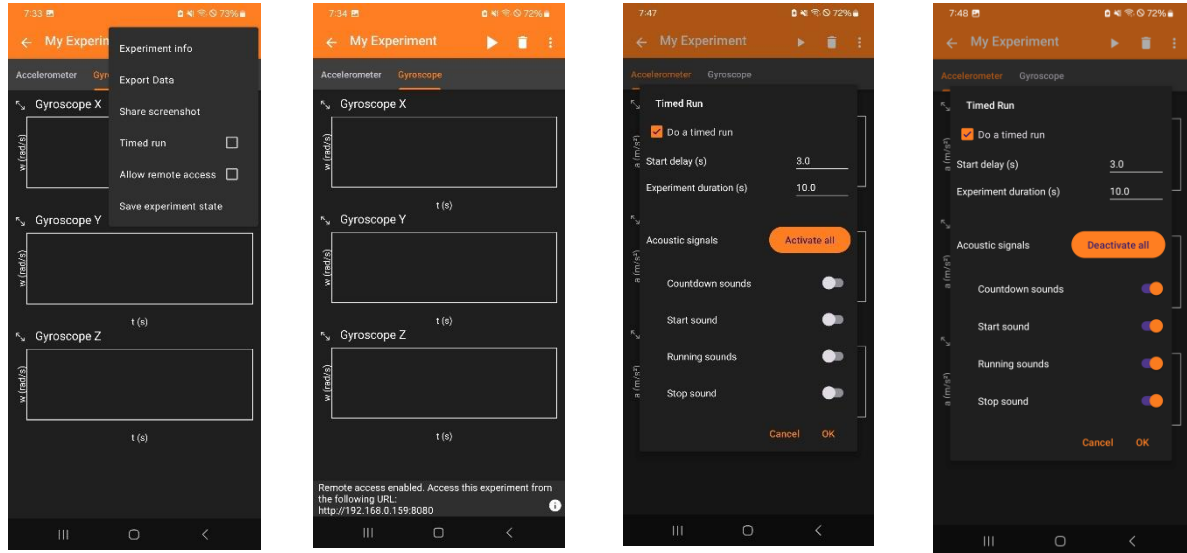


Figure 8 : Enabling remote connect/timed run to prevent interruption in data acquisition

- Remote operation was enabled using the phyphox remote interface (phone hotspot and remote device) so recording could be started/stopped hands-free without disturbing the phone mounting.
- Logging was kept in a single continuous timeline per trial, using Start/Stop controls to clearly separate attempts (with a short idle segment before and after each gesture).
- Data export was done using phyphox Export / Save data as CSV/Excel options as shown above (including the time column and all 6 axes), then transferred to the PC.
- During each trial a visual inspection in the remote device was made. For every teeth click a sudden change in the acceleration is observed, and for every jaw movement, a small wave is observed on the gyroscope.

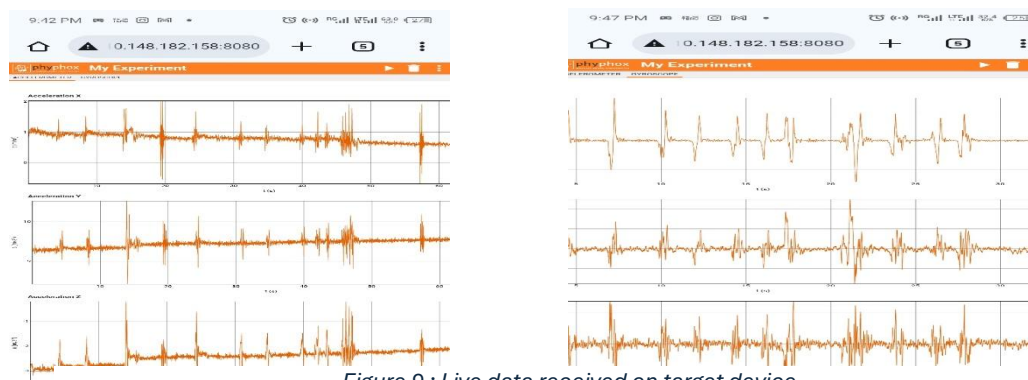


Figure 9 : Live data received on target device

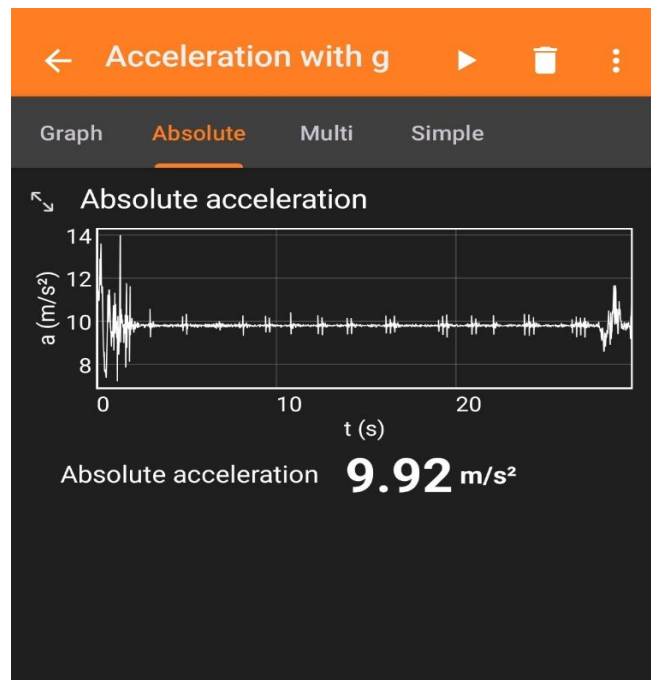
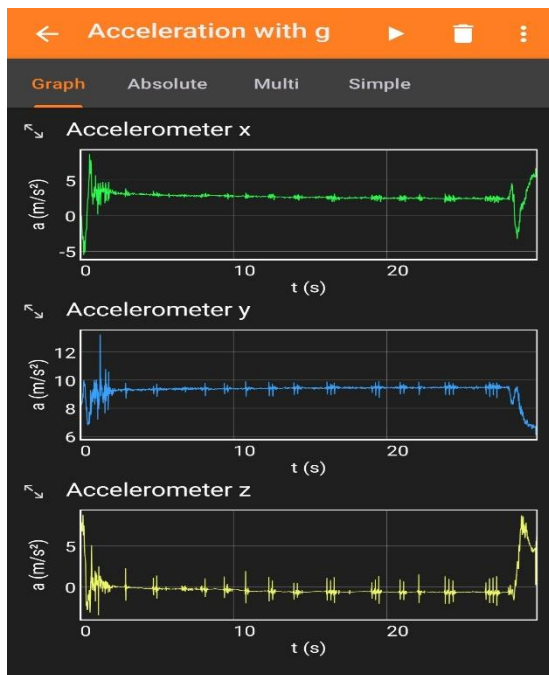


Figure 10 : a. 3 axis Accelerometer time series data, b. Absolute acceleration over time

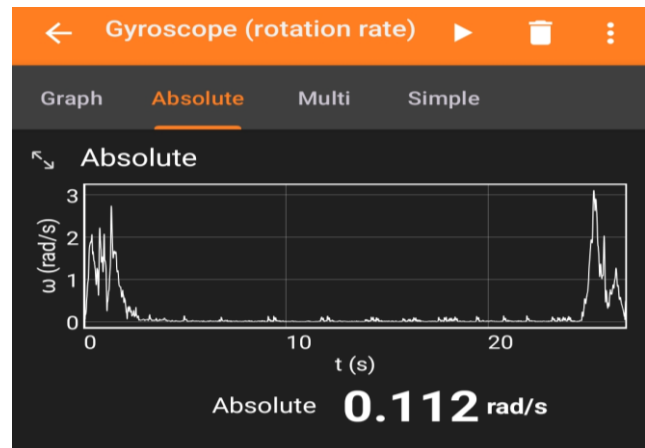
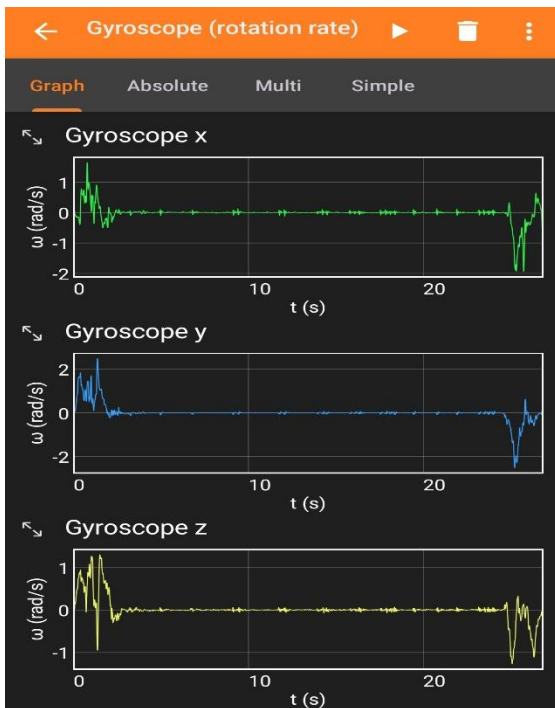


Figure 11 : a. 3 axis Gyroscope time series data, b. Absolute angular velocity over time

3.3 Trials Conducted

The above-said experiment setup is used to obtain the sensor data outputs for different scenarios. Different combinations of vibrations, like tap, tap tap, tap tap tap, cough, and chewing, are made repeatedly with different combinations of head postures, like turning right, turning left, turning up, and turning down, for certain time durations, and the entire time duration is captured. This data specific to the trial combination is then exported as an Excel sheet containing axial information from accelerometer and gyroscope sensors along with the information about the mobile sensors as metadata.

Table 1 : Dataset captured under different conditions

Task	Gesture Target	Noise Present	Head / Motion Condition	Goal
Task 1	Single teeth click	None	Neutral (sitting)	Baseline single-click signature
Task 2	Double teeth click	Chewing	Neutral (sitting)	Double-click under chewing noise
Task 3	Triple teeth click	Background sound	Neutral	Triple-click under ambient noise
Task 4	Teeth click(s)	Low / none	Head turned right	Robustness vs right head rotation
Task 5	Teeth click(s)	Cough + chewing	Head turned left	Click vs combined noise + rotation
Task 6	Teeth click(s)	Cough	Head tilted up	Click vs tilt-up + cough artifact
Task 7	Teeth click(s)	Cough	Head down + up/down transitions	Click vs posture transitions + cough
Task 8	1 / 2 / 3 click sequence	Beat-like impulses	Neutral	Validate timing gaps for click counting
Task 9	Jaw movement + teeth click	Natural jaw motion	Neutral	Separate jaw motion from discrete clicks

The dataset of each task was taken in to PC and articulated under the project root folder “**Data_Sets_acquired**” inside the specific task folder.

4. Preprocessing

Before feature extraction and gesture detection, the exported phyphox datasets were cleaned and standardized. The main goal of preprocessing was to (1) organize inputs/outputs clearly, (2) bring accelerometer and gyroscope datasets into one consistent dataset format, and (3) remove start/end inappropriate artifacts by trimming the same time window for both sensors.

4.1 Folder structure

To keep the workflow reproducible and easy to navigate, the project was organized into four main folders:

- **Data_Sets_acquired/**: Raw Phyphox exports grouped by trial
- **scripts/**: Python scripts used for preprocessing and analysis
- **ARTIFACTS/**: All generated outputs such as trimmed Excel files and plots (.png)
- **Literature Survey/**: Reference papers used

This separation ensures raw data is never overwritten and every generated output is traceable.

4.2 Dataset formatting

Phyphox exports sensor data into Excel workbooks containing separate sheets for each sensor stream. For this experiment, preprocessing assumes the file contains:

- Sheet **Accelerometer** with time series data containing
Time (s), Acceleration x (m/s^2), Acceleration y (m/s^2), Acceleration z (m/s^2)
- Sheet **Gyroscope** with its own data,
Time (s), Gyroscope x (rad/s), Gyroscope y (rad/s), Gyroscope z (rad/s)
- During preprocessing, both sheets are combined into a single clean table (**IMU_Combined**) with simplified column names to support later scripts, after performing the time synchronization and trimming.

Time (s), Accel_X, Accel_Y, Accel_Z, Gyro_X, Gyro_Y, Gyro_Z

This unified format avoids repeatedly handling two separate sheets in later analysis.

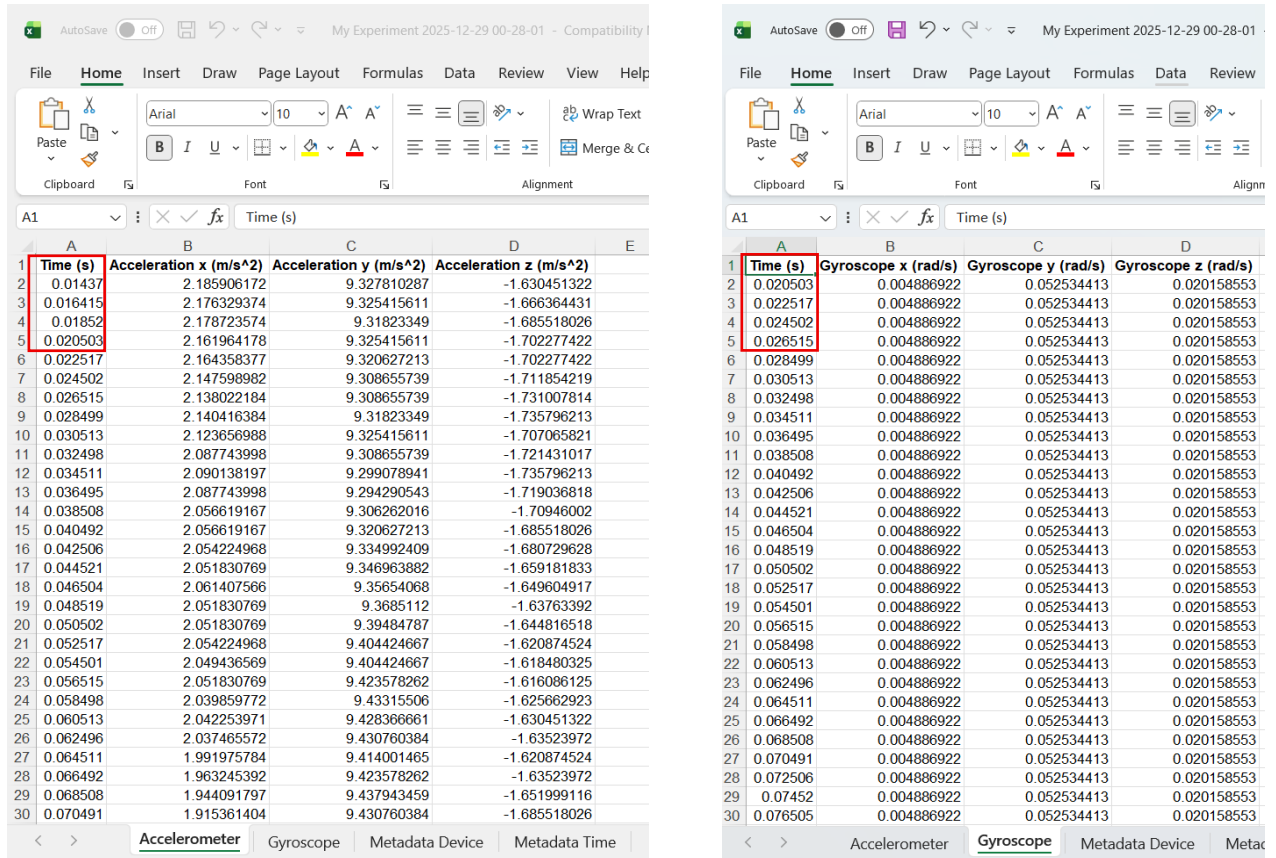


Figure 12 :Different start times of Accelerometer and Gyroscope

- The Metadata sheet gives information about sensor capabilities of the mobile phone used, which indicates the theoretical maximum sampling rate (Min Delay $\sim 2000 \mu s$ 500 Hz), while the effective rate is later verified from the recorded timestamp spacing.

4.3 Time synchronization

As observed from the above image, there is a slight time mismatch between the two sensors. To handle this, the preprocessing script first identifies the common overlapping time interval between the accelerometer and gyroscope recordings:

- Common start time = later of the two start timestamps (in the above case it is 0.020503 S of Gyroscope)
- Common end time = earlier of the two end timestamps

Only this shared interval is kept, ensuring both sensors represent the same real-world time window.

4.4 Trimming

After defining the time overlap, a user-controlled trimming is applied to remove initial and final setup artifacts (default 1 s at the start and 1 s at the end, adjustable via **--start** and **--end**). We can alter the start and end if more time is taken to record the actual gestures beyond the start delay set in the timed run of phyphox application.

Finally, the trimmed accelerometer and gyroscope signals are merged using the matching Time(s) column (inner-join), producing a synchronized dataset where each timestamp contains both Accel (X/Y/Z) and Gyro (X/Y/Z) written into the **IMU_Combined** sheet, which will be available in the **ARTIFACTS** folder.

The name of the file will follow the naming convention mentioned below,

1. <File name from where the excel was read>
2. <Excel name>
3. *_TRIMMED_with_IMU_features* common tag

<Task1_Tooth_Single_Click>_<My_Experiment_2025-12-29_00-28-01>_TRIMMED_with_IMU_features.

5. Feature Engineering

5.1 Implementation methods

After preprocessing, all trimmed trial files stored in the **ARTIFACTS/** folder were used as inputs for the feature engineering and visualization pipeline. The implementation was separated into two Python scripts to keep the workflow modular: one script generates the derived IMU features and updates the dataset, and the second script reads the processed datasets and automatically produces all required plots for the report.

- **Feature computation** (add_features_IMU.py):

Each **_TRIMMED.xlsx* file is loaded, and the **IMU_Combined** sheet is selected as the analysis input. The script computes accelerometer and gyroscope magnitudes, applies the RC high-pass filter ($f_c = 20$ Hz), derives the absolute high-pass envelope (HP_ABS), and calculates rolling-window features such as RMS and SD (and axis-level statistics where applicable). The enriched dataset is then exported back to Excel

as **_TRIMMED_with_IMU_features.xlsx* so every trial contains both raw aligned IMU data and the engineered features in a reproducible format.

- **Plot generation** (Generate_plots_All.py)

The plotting script iterates through all **_with_IMU_features.xlsx* files and generates a standard set of figures per trial (raw axes plots, magnitude plots, HP/HP_ABS traces, and rolling RMS/SD trends). Each plot is saved as .png in the output artifacts folder, enabling consistent visual comparison across single/double/triple click trials and noisy conditions (chewing/coughing/head motion). This automated plotting ensured that all plots required by the assignment were produced in a consistent format without manual editing.

5.2 Steps followed in feature selection

When first plotted the raw accelerometer and gyroscope axes (X/Y/Z) versus time, the click impulses were visible, but they did not appear equally on all axes. In some trials the impulse was dominant on one axis, while in other trials it shifted to a different axis. This made sense because the phone orientation and the strap placement decide how the TMJ vibration projects onto the sensor coordinate frame.

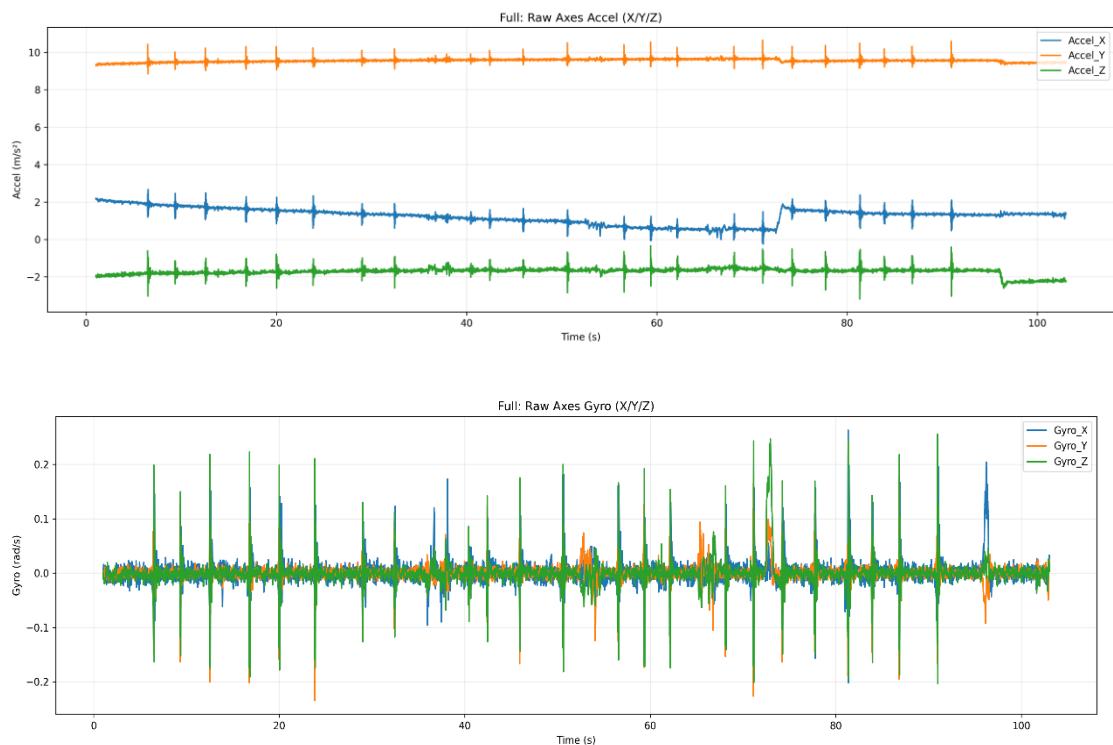


Figure 13 : Raw sensor data for single teeth click

So instead of relying on a single axis (which would make the analysis orientation-dependent), combined the three axes into a **resultant magnitude**. This converts the 3-D motion into one scalar “overall energy” signal and gives a consistent way to compare trials:

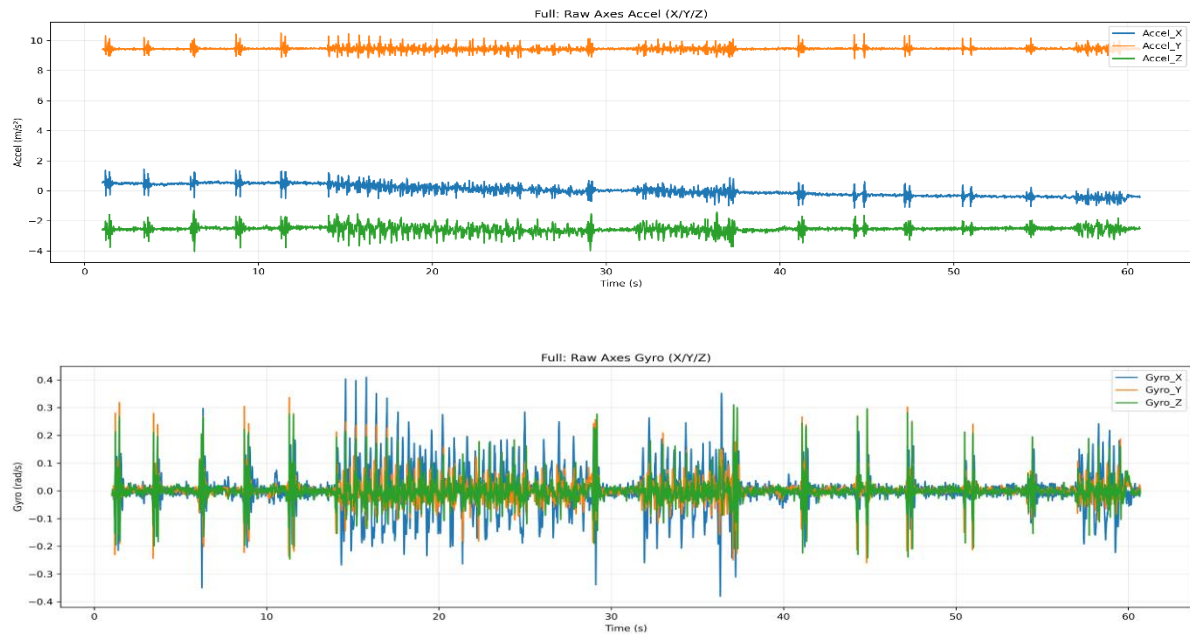


Figure 14 : Raw sensor data for double click with noise

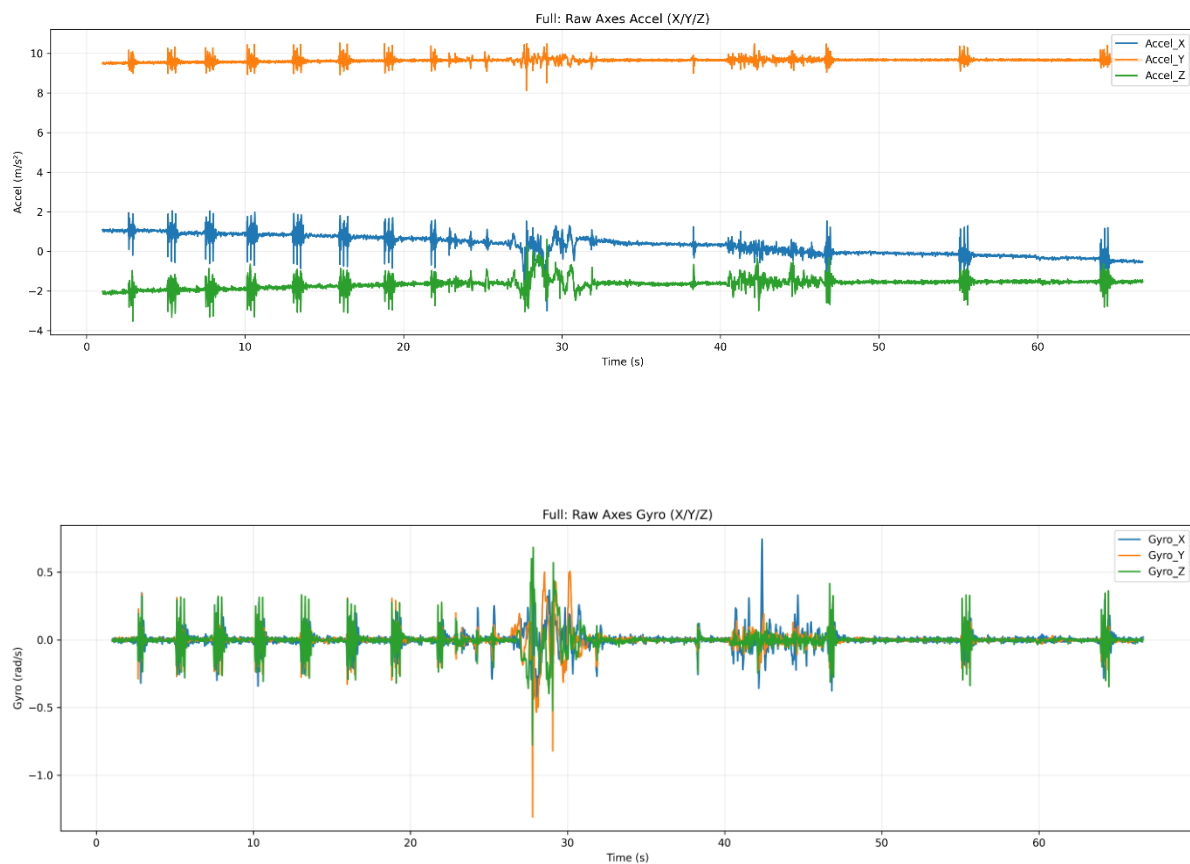


Figure 15 : Raw sensor data for triple click

5.3 Accelerometer and Gyroscope Magnitude

The accelerometer captures linear acceleration, so teeth click (a fast transient vibration) should naturally show up as short, sharp impulses. The gyroscope measures angular velocity, so it is not expected to be highly sensitive to a purely translational shock. However, near the jaw/strap setup, the click can introduce a tiny micro-rotation (e.g., phone rocking on the strap or skin coupling), and that explains why the gyroscope still shows a response during strong clicks. In other words, accelerometer provides the primary “shock/transient” evidence, while the gyroscope can act as supporting evidence that a rotational component was also present.

$$|a| = \sqrt{a_x^2 + a_y^2 + a_z^2},$$

$$|G| = \sqrt{G_x^2 + G_y^2 + G_z^2}$$

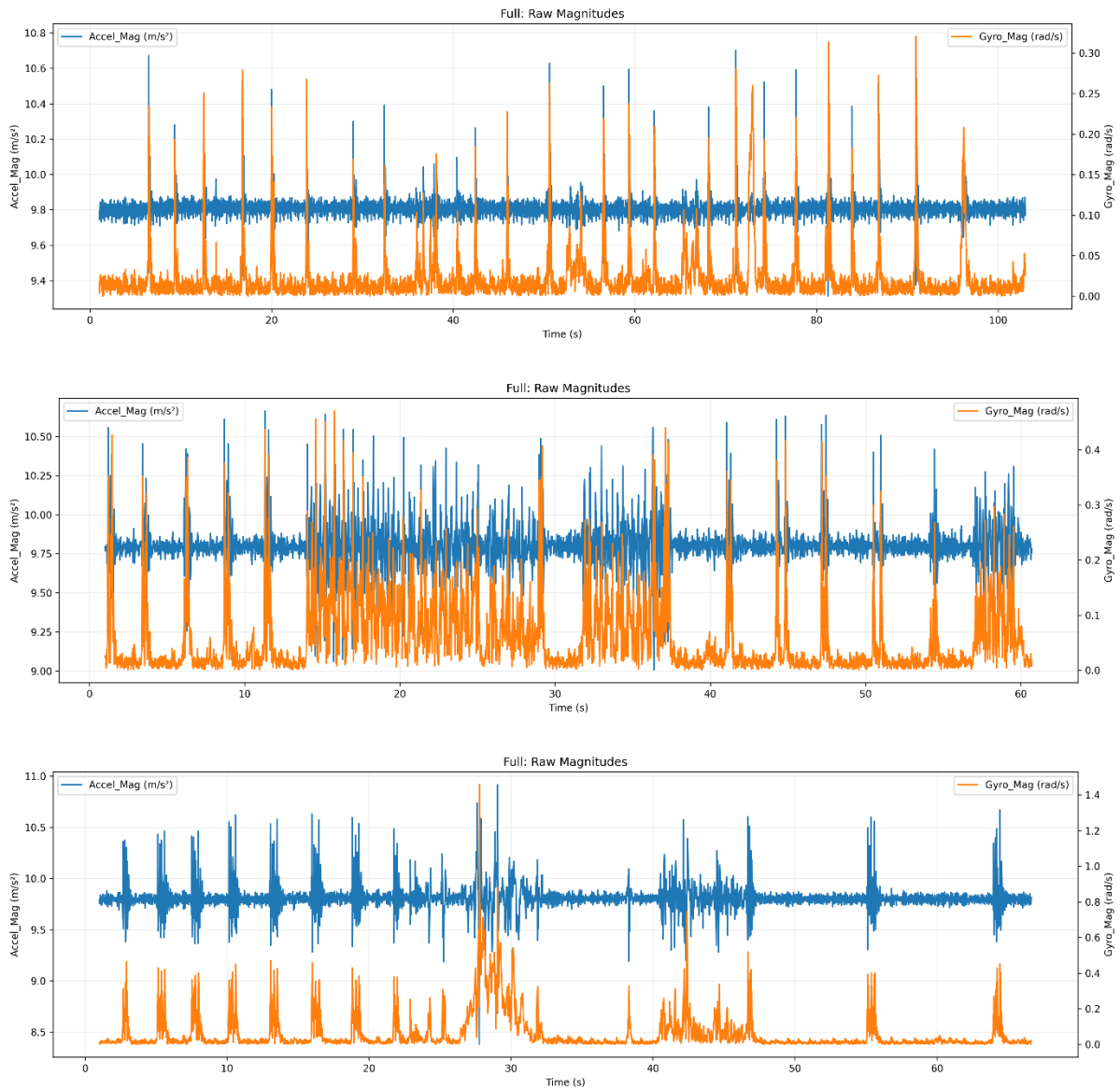


Figure 16 : Accelerometer and Gyroscope Magnitude for single/double/triple tap

On Observing both magnitudes apart from the noisy region, all the clicks have similar magnitude but the time duration was slightly different with single being minimum and triple maximum.

A brief literature review on TMJ click recordings indicates that a “click” is not only an audible phenomenon but also produces measurable mechanical vibrations. Widmalm et al. compared recordings made with microphones in the ear canal versus a skin contact transducer above the joint and reported that the contact sensor retains mainly lower-frequency vibration energy, while higher-frequency components are more prominent in microphone recordings.

This is relevant to the current experiment because a smartphone IMU placed near the TMJ acts more like a contact vibration sensor than an audio microphone. Given the phone’s sampling constraints (≈ 500 Hz), the analysis focuses on capturing and emphasizing the transient mechanical signature of click events rather than the full acoustic spectrum. This motivates using preprocessing/feature steps (e.g., high-pass filtering and windowed energy measures) that highlight short impulses while suppressing slow motion and posture effects.

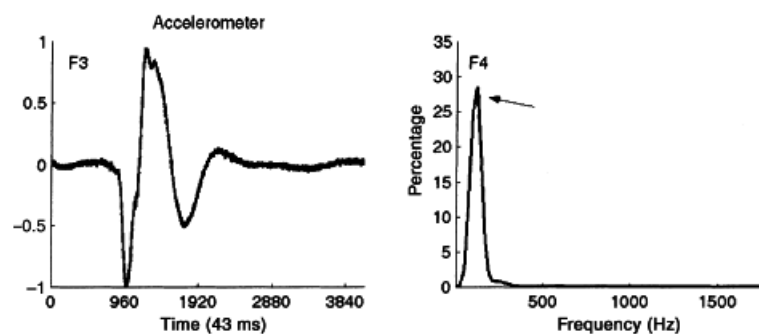


Figure 17: TMJ frequency response recorded by Widmalm et al.

On Observing both the accelerometer magnitude reading, the one recorded with phyphox experiment has the g constant 9.8 m/s^2 which could act as DC and cause noisy output.

5.2 High Pass Filter

To suppress the slow DC/low-frequency component (mainly gravity projection contains $\sim 9.81 \text{ m/s}^2$ gravity baseline which acts as a DC component and can mask short transients.) and emphasize the fast click-like transients, an RC high-pass filter with cutoff frequency = 20 Hz was applied to the IMU signals. With an effective sampling rate of $\sim 496 \text{ Hz}$ (Nyquist $\sim 248 \text{ Hz}$), this keeps the transient vibration band relevant for click detection while removing slow baseline changes.

- $y[n] = \alpha(y[n-1] + x[n] - x[n-1]),$
 $x[n] = \text{input signal}; y[n] = \text{high-pass filtered output}$
- $\alpha = \frac{RC}{RC + \Delta t},$
- $RC = \frac{1}{2\pi f_c},$
- $\Delta t = 1/f_s.$

5.3 High Pass Absolute

The output from HP filter is converted in to a single positive signal to detect the intensity of the signal and to use it as an input to Root Mean Square (RMS) and Standard Deviation (SD) (to find variation) so that the sign-independent envelope is created which is simple for threshold/peak logic and easier to visualize.

$$RMS(x) = \sqrt{\frac{1}{N} \sum x^2} \quad SD(x) = \sqrt{\frac{1}{N} \sum (x - \bar{x})^2}$$

$\bar{x}$: Mean

x : Signal magnitude

N : Number of samples

Rolling RMS was computed as a windowed mean-square amplitude (energy-like envelope) to highlight short transient activity. Click events are then detected by comparing this envelope (or High Pass Absolute peaks) against a robust threshold estimated from baseline statistics (e.g., median and Median Absolute Deviation), which is more stable than using a global mean in the presence of outliers.

In the trials conducted the window size is kept as 200ms which gave approximately 99 samples per window. (Reduce window size if needed the exact location where the click has happened).

On observing the filtered high pass signals for single, double and triple click scenarios the number of peaks and the duration between them clearly varies as expected. The number of peaks increase with increase in clicks; the magnitude of triple click is reduced to half of the magnitude of single click signal while the time interval increased.

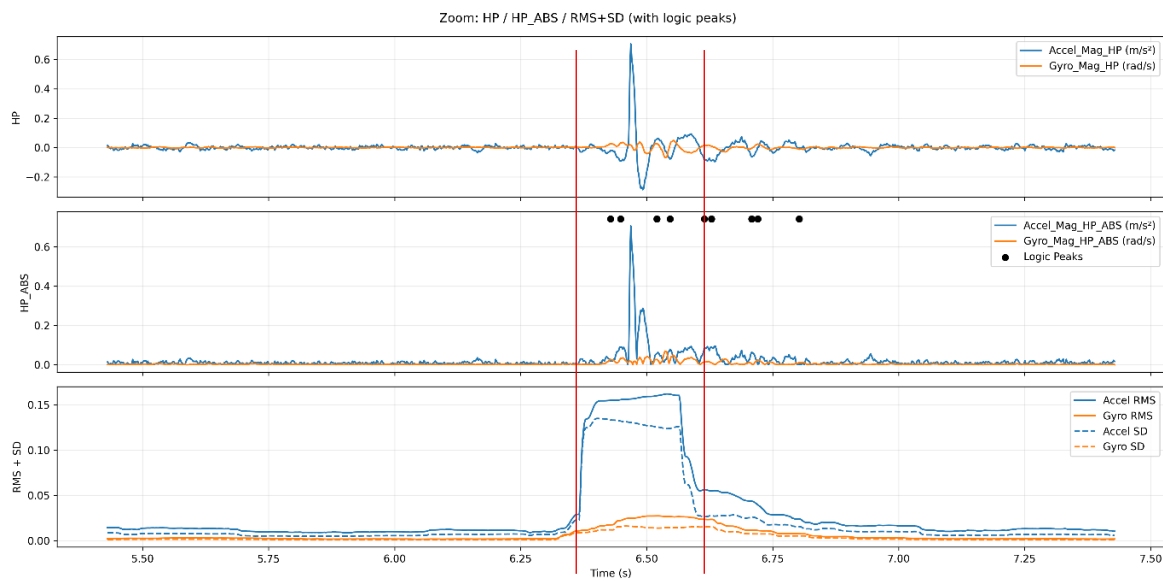


Figure 18 : Filtered single click IMU response with its absolute, RMS & SD

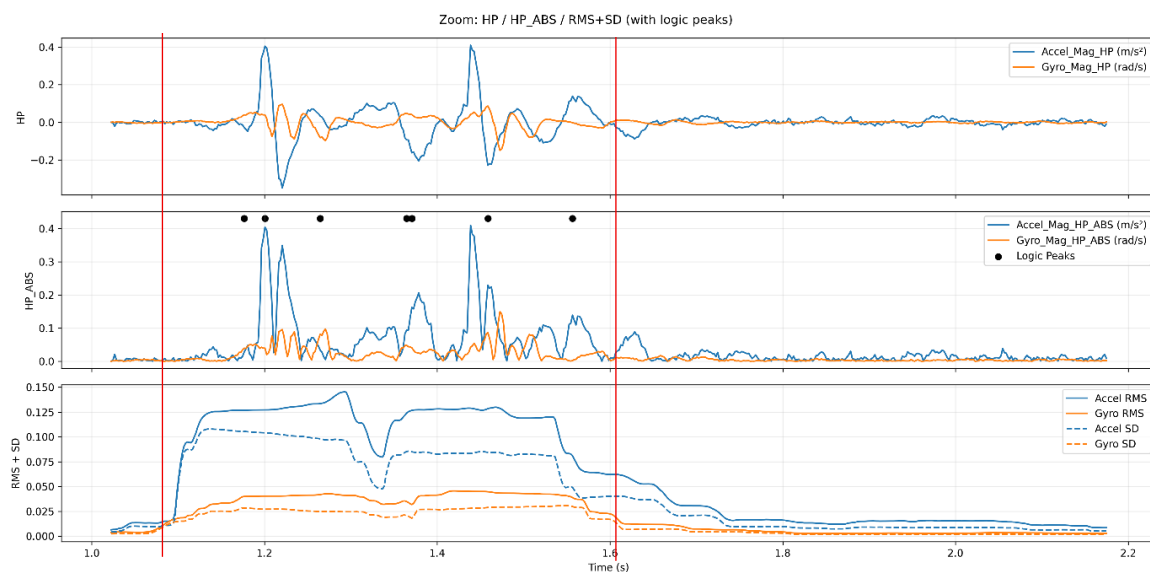


Figure 19 : Filtered double click IMU response with its absolute, RMS & SD

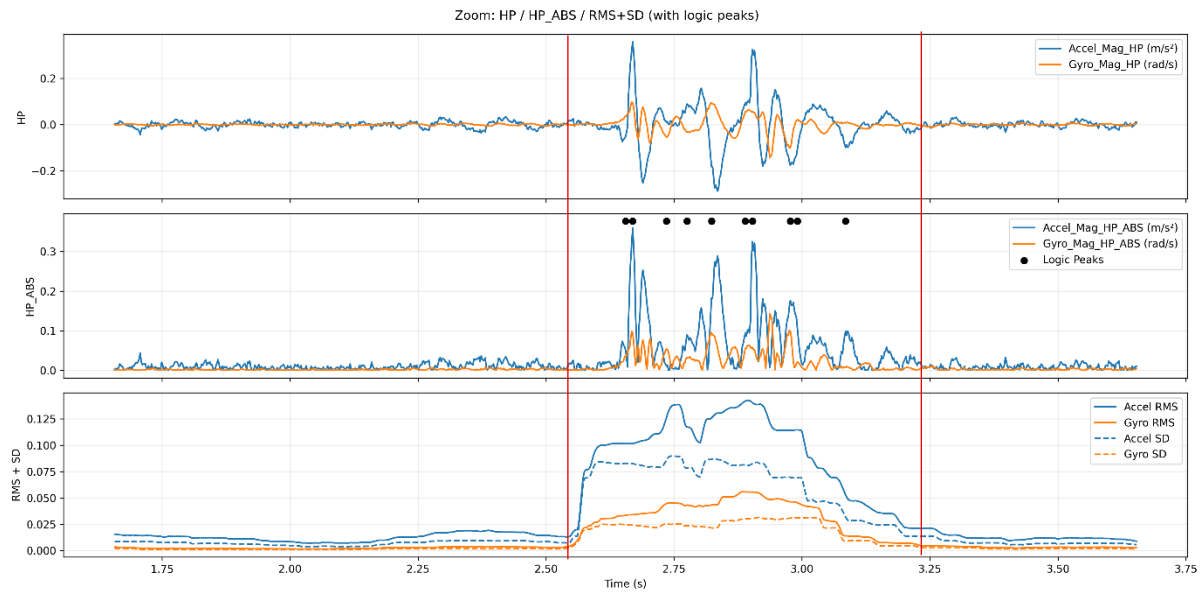


Figure 20 : Filtered triple clicks IMU response with its absolute, RMS & SD

This gives hope to consider each scenario as an individual event and if a proper event condition to confirm it is identified then these gestures can be replicated and used in real times.

5.4 False Positives Identification

Test scenarios for chewing and coughing were generated and when the pattern is tested for the teeth click there is a misinterpretation of chewing as continuous clicking. These false positives were reduced significantly once rolling window method is used to identify the region based peak impulses rather following global threshold.

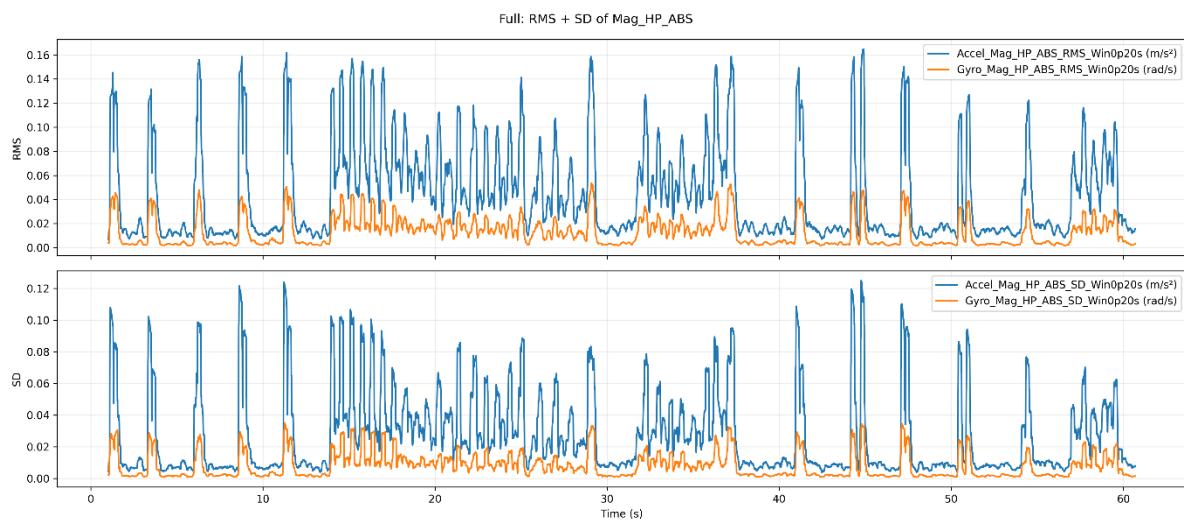


Figure 21 : RMS & SD to detect variation of signal in Noise region

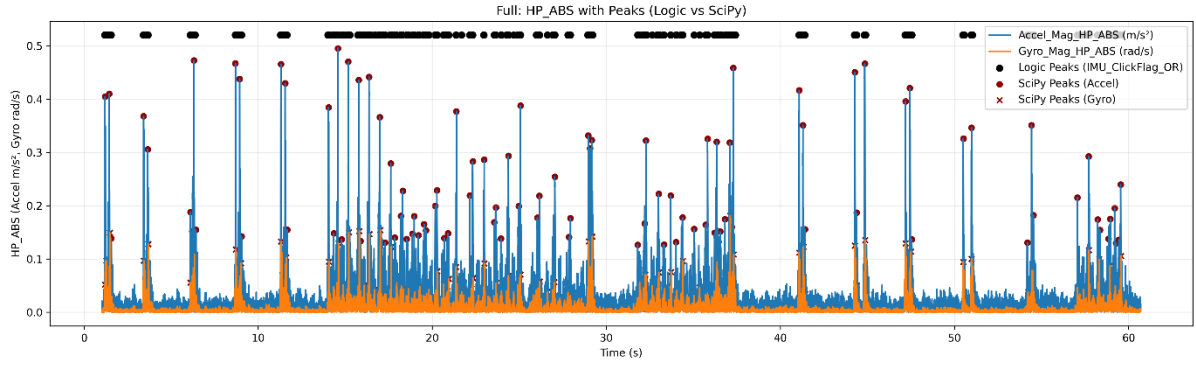


Figure 22 : Peak detection using Global mean as a threshold

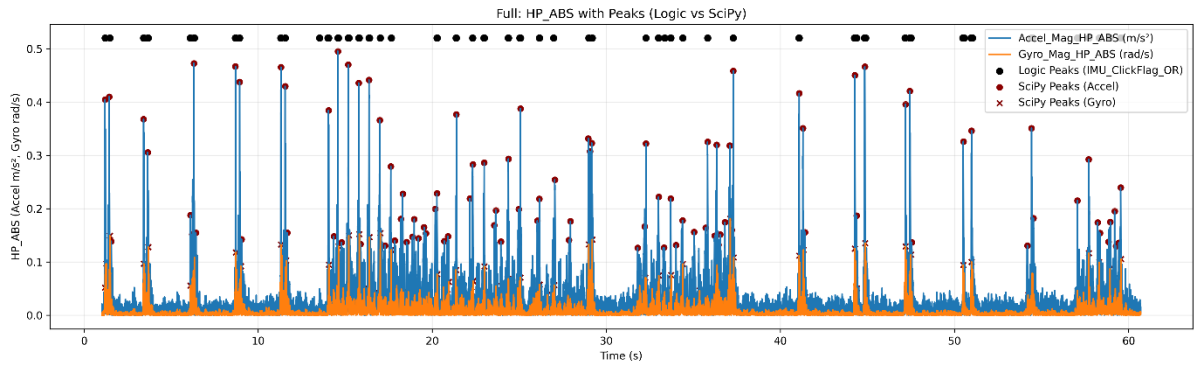


Figure 23 : Peak detection using median absolute deviation as a threshold

For verification of peaks above threshold SciPy peak function is used which significantly matched with the normal time spacing logic. Further jaw movements need to be studied so that angular movements from gyroscope can be usepythond as a flag to detect the chewing and coughing.

6. Conclusion

In this attempt, teeth-click patterns were treated as a physical gesture and verified as measurable IMU events. The raw accelerometer and gyroscope axis plots showed that click impulses are present, and by transforming the signals into a magnitude representation and applying high-pass/envelope features, the click transients became easier to track across trials. Based on these observations, the study supports the idea that single, double, and triple clicks are differentiable, mainly through the number of impulse groups and their time spacing, rather than by magnitude alone. Therefore, future work will focus on improving robustness by (i) fine-tuning the detection sensitivity parameter (k) and the rolling window size to balance missed clicks versus false positives, and (ii) studying gyroscope motion patterns related to jaw movements as a context signal to better reject impulses caused by chewing, coughing, or other non-click events.

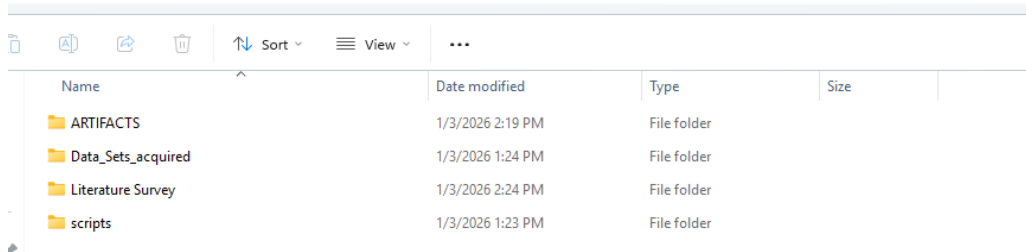
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8. APPENDIX

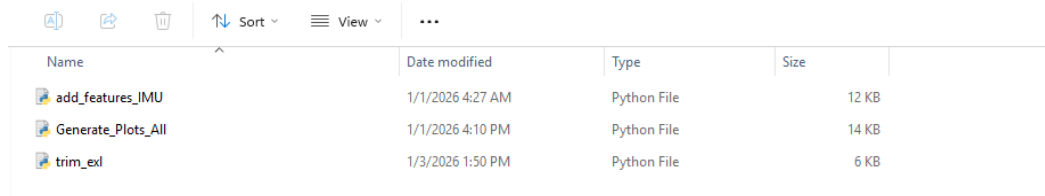
8.1 How to run the scripts

1. Current project folder structure



Name	Date modified	Type	Size
ARTIFACTS	1/3/2026 2:19 PM	File folder	
Data_Sets_acquired	1/3/2026 1:24 PM	File folder	
Literature Survey	1/3/2026 2:24 PM	File folder	
scripts	1/3/2026 1:23 PM	File folder	

2. Scripts folder contains the scripts to process the data and generate plots.



Name	Date modified	Type	Size
add_features_IMU	1/1/2026 4:27 AM	Python File	12 KB
Generate_Plots_All	1/1/2026 4:10 PM	Python File	14 KB
trim_exl	1/3/2026 1:50 PM	Python File	6 KB

3. Run the first command

“python trim_exl.py --input ../Data_Sets_acquired --start 1 --end 1” from the project root folder, so that respective artifacts are generated within ARTIFACTS folder.

```
\Proj_2.0\scripts>python trim_exl.py --input ../Data_Sets_acquired --start 1 --end 1
aw\Bosch\Proj_2.0\scripts\..\ARTIFACTS\Task1_Tooth_Single_Click_My_Experiment_2025-12-29_00-28-01_TRIMMED.xlsx
aw\Bosch\Proj_2.0\scripts\..\ARTIFACTS\Task2_Tooth_Double_Click_Sitting_with_Chew_as_Noise_My_Experiment_2025-12-29_00-41-57_TRIMMED.xlsx
aw\Bosch\Proj_2.0\scripts\..\ARTIFACTS\Task3_Tripe_Clicks_with_Sound_as_Noise_My_Experiment_2025-12-29_00-49-54_TRIMMED.xlsx
aw\Bosch\Proj_2.0\scripts\..\ARTIFACTS\Task4_Turn_Right_Click_My_Experiment_2025-12-29_01-02-21_TRIMMED.xlsx
aw\Bosch\Proj_2.0\scripts\..\ARTIFACTS\Task5_Turn_Left_Click_Cough_Chew_as_Noise_My_Experiment_2025-12-29_01-09-24_TRIMMED.xlsx
aw\Bosch\Proj_2.0\scripts\..\ARTIFACTS\Task6_Turn_UP_Click_Cough_as_Noise_My_Experiment_2025-12-29_01-13-34_TRIMMED.xlsx
aw\Bosch\Proj_2.0\scripts\..\ARTIFACTS\Task7_Turn_Down_Click_Up_Down_Cough_UP_My_Experiment_2025-12-29_01-18-20_TRIMMED.xlsx
aw\Bosch\Proj_2.0\scripts\..\ARTIFACTS\Task8_one_two_three_clicks_with_series_of_beats_My_Experiment_2025-12-29_01-25-03_TRIMMED.xlsx
aw\Bosch\Proj_2.0\scripts\..\ARTIFACTS\Task9_Jaw_movement_and_teeth_clicking_My_Experiment_2025-12-29_02-19-55_TRIMMED.xlsx
```

4. To generate the additional features from IMU axis information run the next command

python add_features_IMU.py --input ../ARTIFACTS --mad_mode rolling

or

python add_features_IMU.py --input ../ARTIFACTS --mad_mode global

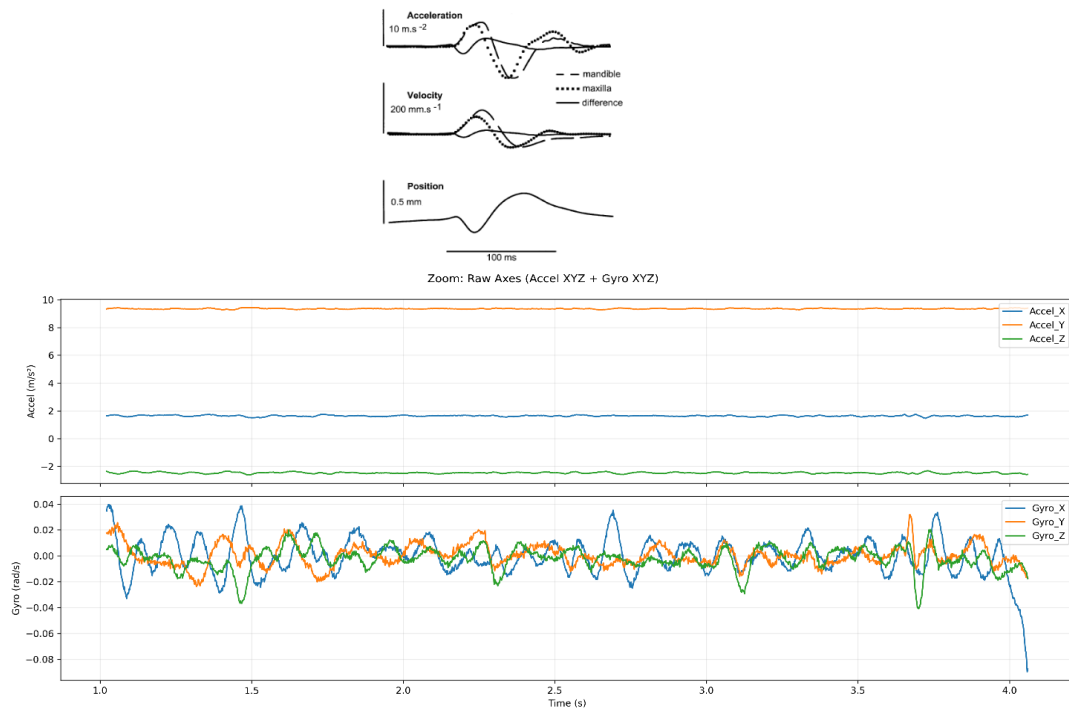
****rolling : less false peak detection.***

5. To generate plots use the command

python Generate_Plots_All.py --artifacts_dir ../ARTIFACTS

8.2 Additional plot

Prior work shows jaw-related mechanical events generate measurable transient inertial signatures under contact coupling; while accelerometers capture the dominant translational shock, gyroscope response can occur when the event produces micro-rotation through jaw hinge motion or strap/skin coupling. This looks similar to gyroscope readings initially, but need to confirm with extensive testing.



8.3 Meta data

Figure 24 : a. Jaw movement signal captured in reference paper, b. Raw gyroscope axial information for jaw tilt

Meta data specific to trial and common for testing are captured in the generated in the ARTFACTS folder in **_with_IMU_features.excel* file under IMU_Feature_Meta sheet, post the features are generated by the *add_features_IMU.py* script.

Figure 25 shows a screenshot of an Excel spreadsheet with the following data:

A	B	C	D	E	F	G	H	I	J	K
fs_hz	HP_FC_HZ	STAT_WIN_S	STAT_WIN_SAMPLES	MAD_MODE	PEAK_K	PEAK_MIN_INTERVAL_S	ACCEL_PEAKS	GYRO_PEAKS	IMU_OR_PEAKS	IMU_AND_PEAKS
496.7890043	20	0.2	99	rolling	3	0.08	140	172	309	3

Figure 25 : Meta data captured